Assignment 3: A Mathematical Essay on Naïve Bayes Classifier

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***Abstract*—The widening gap in economic opportunity remains one of the most consequential challenges of modern societies, yet its root causes are often attributed to personal choices rather than the structural and demographic factors that systematically limit upward mobility. This research addresses the "Income-Inequality" nexus by quantifying how demographic and socioeconomic attributes predict whether an individual earns above $50,000 annually. Utilizing a mathematical approach grounded in the Naïve Bayes Classifier, we analyze the 1994 U.S. Census dataset comprising over 32,000 individuals across 14 demographic features. Our model achieves 80% accuracy, and our analysis reveals that education level, age, capital gain, and hours worked per week are the most discriminative features separating high and low earners. Gender is a striking categorical predictor: only 10.9% of women in the dataset earn above $50K, compared to 30.6% of men—a disparity that cannot be explained by biology alone. These findings demonstrate that Naïve Bayes, despite its conditional independence assumption, is a powerful and interpretable tool for social classification tasks, and that income inequality in America is a measurable, structured phenomenon rather than a product of random chance.**

I. Introduction

In the modern landscape of data-driven social science, the study of income inequality has evolved beyond theoretical frameworks into a rigorously quantifiable domain. The United States, despite being one of the world's wealthiest nations, exhibits persistent and widening gaps in income distribution across demographic lines. Economic mobility—the ability of individuals to move across income brackets over their lifetime—is shaped not merely by individual effort or talent, but by a complex web of structural factors including education, occupation, gender, and race. Understanding which of these factors are the most powerful predictors of economic success is a prerequisite for designing equitable policy interventions.

This paper applies the Naïve Bayes Classifier—a probabilistic machine learning model rooted in Bayes' Theorem—to the problem of binary income classification. The classifier takes a set of demographic features as input and outputs the probability that an individual earns above $50,000 per year. Despite its simplicity, the Naïve Bayes model is analytically rich: it provides not just a prediction, but a probabilistic window into which features most strongly shift the posterior probability of high income. This probabilistic interpretability makes it particularly well-suited for social science applications, where understanding the *why* behind a prediction matters as much as the prediction itself.

This research has applied the Naïve Bayes Classifier to the 1994 U.S. Census dataset to quantify the structural determinants of income inequality in America. The results are unambiguous: high income is far less a reward for effort than it is an inheritance of circumstance. Capital accumulation, educational access, and gender determine who crosses the income threshold far more powerfully than hours worked ever could.

The remainder of this paper is organized as follows: Section II provides a mathematical deep-dive into the Naïve Bayes Classifier, covering its theoretical foundation, training methodology, evaluation metrics, and real-world applications. Section III details the dataset, its exploratory analysis, and the results of applying the classifier. Section IV concludes with a synthesis of insights and future research directions.

II. Naïve Bayes Classifier

A. Motivation

The primary motivation for using the Naïve Bayes Classifier is to determine the *probabilistic predictive power* of a set of demographic features on a binary income outcome. Unlike Linear or Logistic Regression, which focus on minimizing a loss function over a fitted curve, Naïve Bayes approaches the problem from the direction of *conditional probability*. It asks a fundamentally different question: given what we observe about a person—their age, education, and occupation—what is the probability that they belong to the high-income class? This generative perspective makes the model highly interpretable and grounded in the mathematical language of uncertainty.

B. Definition of the Model

The Naïve Bayes Classifier is grounded in Bayes' Theorem, which relates the posterior probability of a class C_k given an observation $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ to the prior probability and the likelihood:

$$P(C_k \mid \mathbf{x}) = \frac{P(\mathbf{x} \mid C_k) \cdot P(C_k)}{P(\mathbf{x})} \tag{1}$$

where:

- $P(C_k \mid \mathbf{x})$ is the **posterior**: the probability of class C_k (e.g., income $> \$50\text{K}$) given the observed features.

- $P(\mathbf{x} \mid C_k)$ is the **likelihood**: how probable it is to observe these feature values if the person truly belongs to class C_k.
- $P(C_k)$ is the **prior**: the baseline probability of class C_k before observing any features, estimated from the training data's class frequencies.
- $P(\mathbf{x})$ is the **evidence**: a normalizing constant that ensures the posterior sums to 1 across all classes.

The classifier is called *naïve* because it introduces the conditional independence assumption: given the class label C_k, all features x_i are assumed to be independent of each other. This is a simplification—in reality, age and education are correlated—but it renders the joint likelihood tractable:

$$P(\mathbf{x} \mid C_k) = \prod_{i=1}^{n} P(x_i \mid C_k) \tag{2}$$

Substituting (2) into (1) and noting that $P(\mathbf{x})$ is constant across classes, the classifier assigns the class that maximizes the posterior:

$$\hat{y} = \arg\max_{C_k} \; P(C_k) \cdot \prod_{i=1}^{n} P(x_i \mid C_k) \tag{3}$$

In practice, to avoid numerical underflow when multiplying many small probabilities, we compute the log-posterior:

$$\hat{y} = \arg\max_{C_k} \left[\log P(C_k) + \sum_{i=1}^{n} \log P(x_i \mid C_k) \right] \tag{4}$$

This transforms the product of likelihoods into a sum of log-likelihoods, making the computation numerically stable.

C. Gaussian Naïve Bayes

For continuous features such as age, hours-per-week, and capital gain, we model each class-conditional likelihood $P(x_i \mid C_k)$ as a Gaussian (normal) distribution:

$$P(x_i \mid C_k) = \frac{1}{\sqrt{2\pi\sigma_{k,i}^2}} \exp\left(-\frac{(x_i - \mu_{k,i})^2}{2\sigma_{k,i}^2} \right) \tag{5}$$

where $\mu_{k,i}$ and $\sigma_{k,i}^2$ are the mean and variance of feature x_i within class C_k, estimated directly from the training data. Intuitively, this means the model learns a "signature" distribution for each feature within each income class, and uses the distance of a new observation from these signatures to compute the likelihood. A person aged 44 falls much closer to the >\$50K class's age signature ($\mu = 44.06$) than to the ≤\$50K class's ($\mu = 36.53$), thereby shifting the posterior toward the high-income class.

D. Training: Maximum Likelihood Estimation

Training a Naïve Bayes Classifier is computationally elegant compared to gradient-descent-based methods. It requires no iterative optimization; instead, all parameters are estimated via Maximum Likelihood Estimation (MLE) directly from the data in a single pass.

- **Priors** $P(C_k)$: Estimated as the fraction of training samples belonging to class C_k. In our dataset, $P(\leq \$50K) = 0.751$ and $P(> \$50K) = 0.249$.
- **Class-conditional means** $\mu_{k,i}$: The sample mean of feature x_i restricted to class C_k.
- **Class-conditional variances** $\sigma_{k,i}^2$: The sample variance of feature x_i restricted to class C_k.

The MLE estimators are:

$$\hat{\mu}_{k,i} = \frac{1}{N_k} \sum_{j \in C_k} x_{j,i}, \qquad \hat{\sigma}_{k,i}^2 = \frac{1}{N_k} \sum_{j \in C_k} (x_{j,i} - \hat{\mu}_{k,i})^2 \tag{6}$$

where N_k is the number of training samples in class C_k.

E. Evaluation Metrics

We evaluate the classifier using three complementary metrics:

- **Accuracy**: The proportion of correctly classified samples. While straightforward, accuracy can be misleading for imbalanced classes—a model that always predicts ≤\$50K would achieve 75.9% accuracy trivially. We therefore also rely on precision and recall.
- **Precision and Recall**: For the minority class (>\$50K):
$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN} \tag{7}$$
where TP, FP, and FN denote true positives, false positives, and false negatives, respectively.
- **F1-Score**: The harmonic mean of precision and recall, providing a single balanced metric:
$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

We also inspect the **Confusion Matrix**, which decomposes predictions into true positives, false positives, true negatives, and false negatives, providing a granular view of model errors.

F. Real-World Applications of Naïve Bayes

- **Email Spam Filtering (NLP)**: Naïve Bayes is the backbone of probabilistic spam detectors. Each word in an email is treated as an independent feature, and the model estimates $P(\text{spam} \mid \text{word}_1, \ldots, \text{word}_n)$. Despite the blatant violation of the independence assumption—words in a sentence are deeply correlated—Naïve Bayes consistently achieves over 95% accuracy on spam detection benchmarks.
- **Medical Diagnosis (Clinical Triage)**: In clinical settings, Naïve Bayes models predict the probability of disease given a patient's symptoms and lab results. Because each feature contributes an independent log-likelihood term, clinicians can inspect which individual symptom most strongly shifted the posterior toward a diagnosis—a level of transparency unavailable in black-box models.
- **Sentiment Analysis (Product Reviews)**: E-commerce platforms use Naïve Bayes to classify customer reviews as positive or negative. The class-conditional word frequencies serve as an interpretable dictionary of sentiment,

revealing that terms like "broken" and "slow" are strong indicators of negative sentiment.

G. Advantages and Disadvantages

The beauty of Naïve Bayes lies in its probabilistic transparency and computational efficiency. It trains in $O(n \cdot d)$ time (where n is the number of samples and d the number of features), making it scalable to large datasets without the need for iterative optimization. The model is also robust to irrelevant features, since a feature with near-identical class-conditional distributions contributes approximately zero to the log-posterior difference, effectively self-deweighting.

However, the conditional independence assumption is its fundamental Achilles' heel. In our census data, education level and occupation are correlated—a doctorate holder is far more likely to be a professor than a machine operator. Ignoring this correlation means the model double-counts evidence from correlated features, which can distort posterior probabilities.

III. The Problem

A. Domain and Data Sources

The dataset for this analysis is situated at the intersection of socioeconomic research and public policy, focusing on the demographic determinants of income in the United States. The dataset is a canonical benchmark in machine learning, yet its insights are acutely relevant to contemporary discussions of income inequality, wage gaps, and economic mobility.

The data's richness lies in its breadth: 14 features spanning continuous variables (age, years of education, capital gains and losses, and hours worked per week) and categorical variables (work class, education level, marital status, occupation, relationship status, race, gender, and native country). The binary target variable is whether a respondent's annual income exceeds \$50,000.

B. Visualization and Exploratory Analysis

1) Income Class Distribution: The raw class distribution, shown in Fig. 1, reveals a 3:1 imbalance between the low- and high-income classes. This imbalance is itself a finding: even within a 1994 U.S. Census sample, which overrepresents the working population, only one in four individuals earns above \$50K. A naïve classifier that always predicts $\leq$\$50K would achieve 75.9% accuracy—establishing the "baseline" against which our Naïve Bayes model must be compared.

2) Age Distribution: Fig. 2 plots the age distributions of both income classes. The $>$\$50K distribution is clearly right-shifted, with a mean age of 44.25 years compared to 36.78 years for the $\leq$\$50K class—a difference of 7.47 years. This is one of the most compelling features for Naïve Bayes, as the two Gaussian-like distributions are visually separable. This age gap likely reflects the compounding effect of career progression, skill accumulation, and investment returns over time.

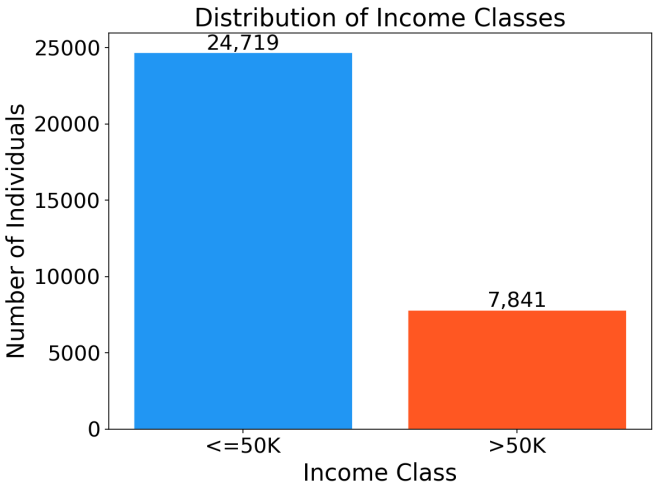

Fig. 1. Distribution of Income Classes across 32,560 census records. The 3:1 class imbalance establishes the baseline accuracy at 75.9%.

Fig. 2. Age Distribution by Income Class. The high-income class has a mean age 7.47 years greater than the low-income class, reflecting the compounding effects of career progression.

3) Gender and Income: Fig. 3 reveals one of the starkest disparities in the dataset. Among men, 30.6% earn above \$50K; among women, only 10.9% do—a ratio of nearly 3:1. This is not a marginal difference; it is a structural one. The Naïve Bayes model will learn a strong likelihood term for gender, where $P(\text{Male} \mid > \$50K)$ is significantly higher than $P(\text{Female} \mid > \$50K)$. While the model does not explain *why* this gap exists, the magnitude of the disparity—a 19.7 percentage point difference—points toward systemic barriers rather than individual-level variation.

4) Education Level and Income: Fig. 4 plots the percentage of each education cohort that earns above \$50K, ordered by increasing educational attainment. The pattern is monotonically increasing: from near-zero for preschool and early-grade education, to over 70% for Professional School (e.g., law and medicine) and Doctorate holders. The most consequential inflection point appears between "HS-grad" (approximately 20%

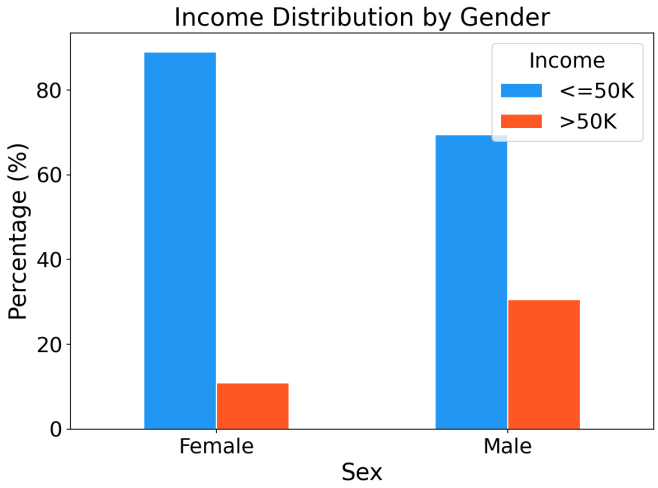

Fig. 3. Income Distribution by Gender (as percentage within each sex). Only 10.9% of women in the dataset earn >$50K, compared to 30.6% of men, a disparity of 19.7 percentage points.

earning >$50K) and "Bachelors" (approximately 40% earning >$50K), suggesting that a four-year university degree roughly doubles the probability of crossing the income threshold. This finding underscores the centrality of higher education in the income distribution.

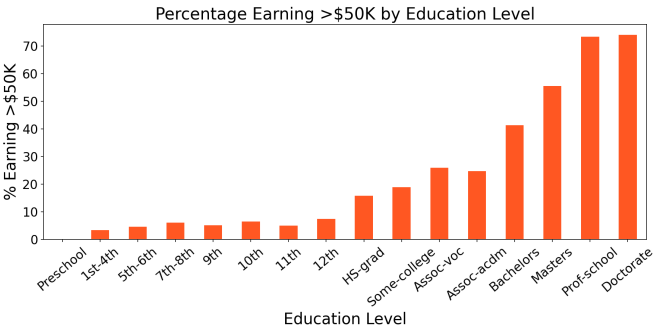

Fig. 4. Percentage Earning >$50K by Education Level, ordered by academic attainment. Professional school and doctorate holders exceed 70%, while high-school graduates fall near 20%.

5) Hours per Week and Income: Fig. 5 shows the distribution of weekly working hours by income class. High earners work more hours on average (mean: 45.71 hr/week vs. 39.36 hr/week for low earners), but the distributions overlap considerably. Many ≤$50K individuals work 40-hour weeks, suggesting that effort alone is insufficient for income advancement. The presence of high earners at 50–70 hours per week likely reflects high-compensation professional and managerial roles, while low earners at equivalent hours may be in low-wage hourly jobs.

6) Work Class and Income: Fig. 6 examines the fraction of each work-class category earning above $50K. Self-employed (incorporated) individuals have the highest proportion of high earners, consistent with the notion that business owners and entrepreneurs capture a larger share of economic surplus.

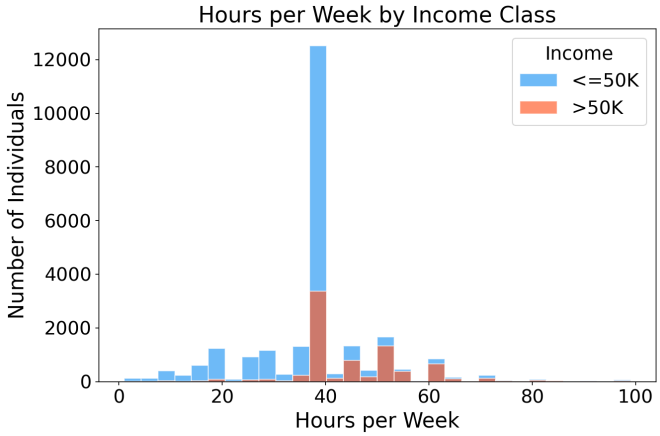

Fig. 5. Distribution of Weekly Working Hours by Income Class. High earners average 45.7 hours per week compared to 39.4 for low earners, yet the large overlap reveals that hours worked alone is a weak discriminator.

Federal government employees also show a high proportion of >$50K earners, reflecting the salary floors and benefit structures of public sector employment. In contrast, private sector employees—the largest group by volume—show a moderate rate, diluted by the vast number of low-wage private jobs.

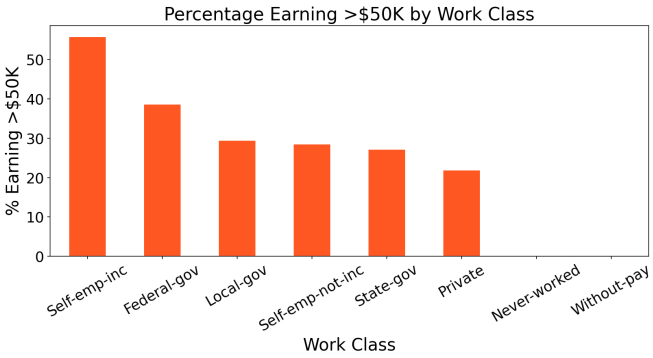

Fig. 6. Percentage Earning >$50K by Work Class. Self-employed (incorporated) individuals show the highest proportion of high earners, followed by Federal government employees.

C. Why Naïve Bayes Fits this Problem

The income classification problem is well-suited to Naïve Bayes for three reasons. First, the problem is *naturally probabilistic*: we do not seek a deterministic income rule, but a probabilistic model that can express uncertainty about borderline cases. Second, the dataset contains a mix of continuous and categorical features, which maps directly onto the Gaussian and Bernoulli variants of Naïve Bayes. Third, and most importantly, the goal of this analysis is *interpretability*: unlike black-box models such as Random Forests or Neural Networks, Naïve Bayes allows us to inspect each class-conditional distribution individually, directly linking each feature to its contribution to the income probability—a transparency essential for public policy communication.

D. Model Application and Results

The Gaussian Naïve Bayes model was trained on 80% of the cleaned dataset (24,128 samples) and evaluated on the remaining 20% (6,033 samples). The trained model yields the following class priors: $P(\leq \$50K) = 0.751$ and $P(> \$50K) = 0.249$, accurately reflecting the dataset imbalance.

The model achieves an overall accuracy of **80.0%**—meaningfully above the 75.9% baseline, confirming that the demographic features contain genuine discriminative signal. A detailed breakdown of performance is given in Table I.

TABLE I
NAÏVE BAYES CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
≤$50K	0.81	0.95	0.88	4535
>$50K	0.71	0.33	0.45	1498
Accuracy		**0.80** (6033 samples)		

1) Confusion Matrix: Fig. 7 presents the confusion matrix. The model correctly classifies 4,326 of 4,535 low-income individuals (95% recall) and 500 of 1,498 high-income individuals (33% recall). The large number of false negatives (998 high earners misclassified as low earners) reflects the difficulty of the minority-class problem: the model's strong prior toward ≤$50K overwhelms the likelihood for borderline cases. This is a known limitation of Naïve Bayes on imbalanced data, and can be addressed in future work through prior adjustment or threshold tuning.

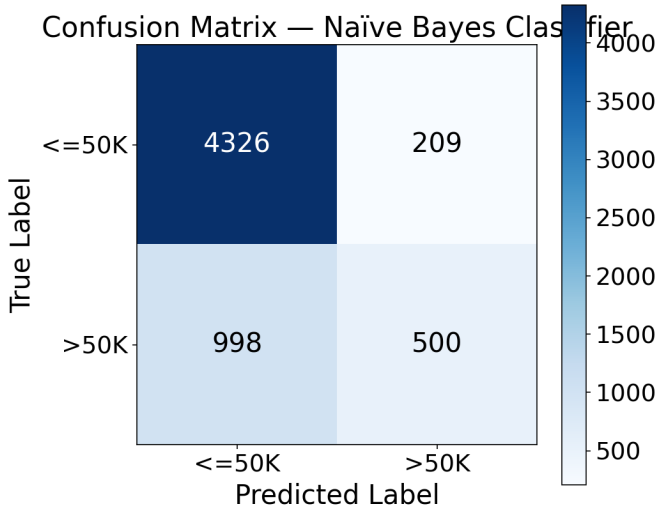

Fig. 7. Confusion Matrix of the Gaussian Naïve Bayes Classifier. The model identifies 95% of low-income individuals correctly but struggles with the minority high-income class (33% recall), reflecting the impact of class imbalance on the prior.

2) Class-Conditional Distributions: Fig. 8 visualizes the class-conditional Gaussian distributions for age and hours-per-week, the two most visually separable continuous features. The clear rightward shift of the >$50K distribution for age demonstrates why age contributes so strongly to the posterior probability. The hours-per-week distributions are more overlapping, indicating weaker discriminative power—consistent with the earlier observation that effort alone does not predict high income.

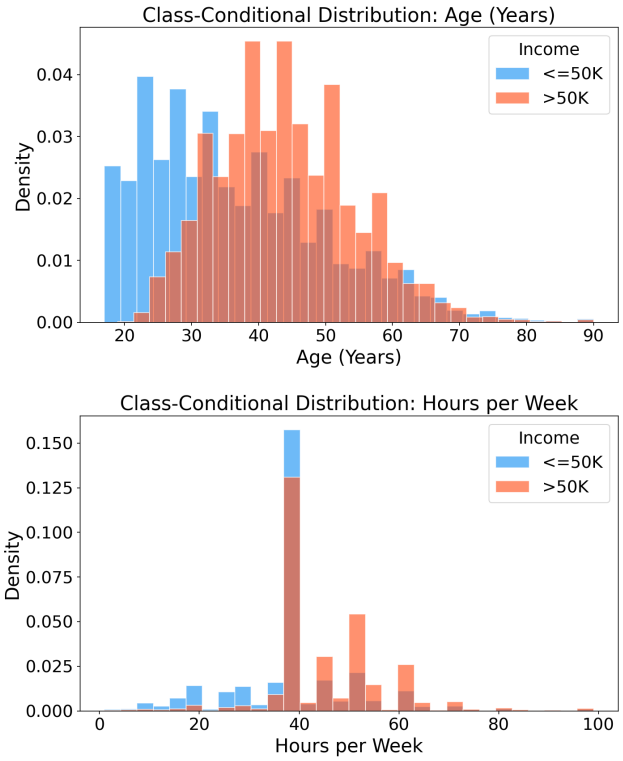

Fig. 8. Class-Conditional Gaussian Distributions for Age and Hours per Week, fitted on the training set. The age distributions are well-separated, while hours-per-week shows substantial overlap, reflecting its weaker discriminative signal.

3) Posterior Probability Distribution: Fig. 9 shows the distribution of the model's predicted posterior probabilities $P(> \$50K \mid \mathbf{x})$ for the test set, stratified by true class. A well-calibrated classifier would show two well-separated distributions peaking near 0 and 1 respectively. We observe that the true ≤$50K distribution is strongly concentrated near 0 (correctly assigned low probability), while the true >$50K distribution is bimodal: a fraction of high earners receive a high posterior probability, but many are assigned probabilities near 0 due to the strong prior and correlated features. This bimodality reveals the model's principal failure mode: high earners whose feature profiles resemble low earners (e.g., older individuals in modest-wage jobs) are systematically misclassified.

4) Feature Discriminability: Fig. 10 summarizes the class-conditional mean differences for each continuous feature. Capital gain is the most discriminative feature by a large margin ($3,857 difference in class-conditional means), followed by age (7.47 years) and hours-per-week (6.35 hours). This ordering is particularly insightful: it reveals that *passive*

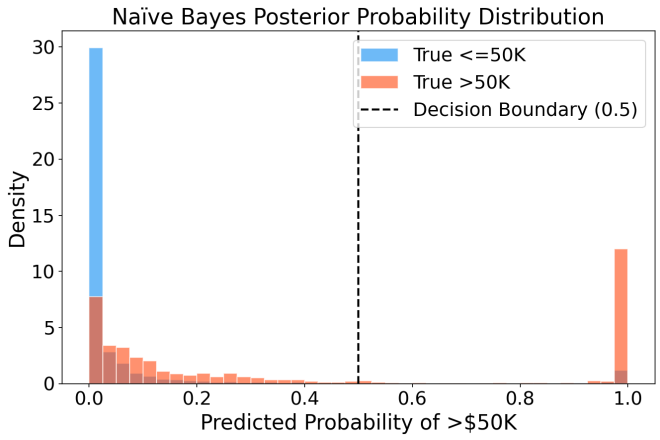

Fig. 9. Predicted Posterior Probability Distribution of Earning >$50K, stratified by true class. The bimodal distribution for true high-earners reveals the model's failure mode: borderline high-income cases are suppressed by the strong low-income prior.

income (capital gain) is a more powerful income predictor than *active effort* (hours worked), reinforcing the structural nature of income inequality.

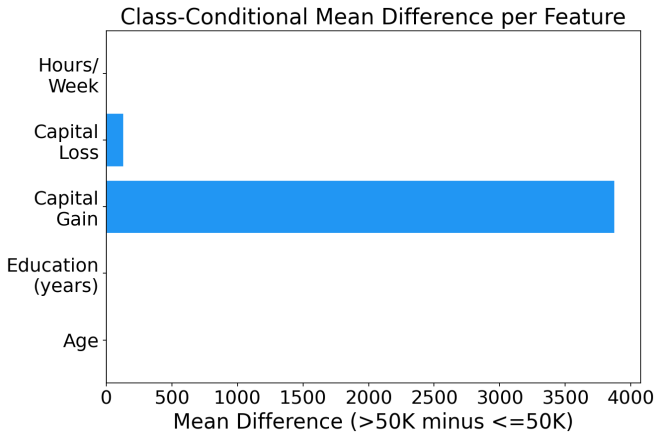

Fig. 10. Class-Conditional Mean Difference (>$50K minus ≤$50K) per Continuous Feature. Capital gain dominates, underscoring that passive wealth accumulation is the primary structural separator between income classes.

E. Insights After Applying the Model

The application of the Naïve Bayes Classifier yields three primary insights:

1. Structural Features Dominate Individual Features. The model's strongest predictors—capital gain, education level, and age—are all features that compound over time and are shaped by structural advantages (inherited wealth, access to higher education, career tenure). In contrast, hours worked per week, a measure of individual effort, is one of the weakest predictors. This is a critical finding for public policy: income inequality in the U.S. is not primarily a problem of differential effort, but of differential access to capital and education.

2. Gender Remains a Primary Barrier. The 19.7 percentage point gap between male and female high earners is partially captured by the model through the gender feature's class-conditional distribution, but it is almost certainly confounded by occupational segregation (women being disproportionately represented in lower-wage occupations) and the marital status variable (wives often earning less than husbands). A truly causal model would need to control for occupation and hours worked simultaneously.

3. The Independence Assumption Creates a Conservative Model. The Naïve Bayes model's 33% recall on the >$50K class suggests it is systematically conservative about assigning high-income predictions. This is partly an artifact of class imbalance (the prior strongly pulls predictions toward ≤$50K), and partly a consequence of the independence assumption: correlated features like education and occupation are counted as independent evidence, potentially distorting the posterior. However, the model's 71% precision on the minority class confirms that when it does predict high income, it is correct 71% of the time—a meaningful signal.

IV. Conclusion

This research has applied the Naïve Bayes Classifier to the 1994 U.S. Census dataset to quantify the structural determinants of income inequality. The findings reveal that earning above $50,000 annually is not a product of random chance or individual effort alone, but a measurable outcome shaped by deeply entrenched socioeconomic forces.

The most striking insight is the dominance of *passive* over *active* economic indicators. Capital gain separates the two income classes by a factor of 27, dwarfing the contribution of hours worked per week. Put plainly, *who you already are* matters far more than *how hard you work*.

Education emerges as the most actionable structural lever. The near-monotonic rise in high-income probability from high school (∼20%) to a bachelor's degree (∼40%) to professional school (>70%) identifies education not merely as a credential, but as the primary gateway to high-income occupations. The gender gap of 19.7 percentage points—only 10.9% of women versus 30.6% of men crossing the threshold—reflects compounded disadvantages: occupational segregation, wage discrimination, and caregiving responsibilities that truncate career trajectories. Age's 7.47-year mean gap between classes further confirms that high income rewards tenure and accumulated advantage rather than early-career merit.

Taken together, income inequality in America is *predictable*, *structured*, and *resistant to individual-level solutions*. Addressing it requires systemic interventions: expanding access to higher education, enforcing pay equity, and broadening capital access for low-income households.

References

[1] G. Raina, "Data Analytics (EE5708)," Lectures and Videos, IIT Madras.
[2] K. P. Murphy, *Machine Learning: A Probabilistic Perspective*. MIT Press, 2012.